CIE AS PHYSICS (9702) • MARK SCHEME & MODEL ANSWERS

# 1.1 Physical Quantities

**Syllabus Section:** 1. Physical Quantities & Units **Paper Type:** Structured Questions (Paper 2) **Total:** 8 Questions (59 Marks)

## Question 1 Easy [4 marks]

### Part (a) [1 mark]

State what is meant by a physical quantity.

**Mark Scheme (1 marks):**

A physical quantity is:

- A property that can be measured

  **OR**

- Something that consists of a magnitude / numerical value and a unit; *[1 mark]*

***[Total: 1 mark]***

### Part (b) [1 mark]

State **one** physical quantity that can be represented by the symbol $\rho$.

**Mark Scheme (1 marks):**

A physical quantity that can be represented by the symbol $\rho$ is:

- Density**OR**
- Resistivity; *[1 mark]*

***[Total: 1 mark]***

### Part (c) [1 mark]

State **one** physical quantity that can be represented by the capital letter $P$ or the lowercase letter $p$.

**Mark Scheme (1 marks):**

A physical quantity that can be represented by $P$ or $p$ is:

Any **one** from:

- Pressure; *[1 mark]*
- Momentum; *[1 mark]*
- Power; *[1 mark]*

***[Total: 1 mark]***

### Part (d) [1 mark]

Explain how the quantities represented by $\rho$, $p$, or $P$ can be distinguished from each other.

**Mark Scheme (1 marks):**

The quantities represented by $\rho$, $p$, or $P$ can be distinguished from each other by:

Any **one** from:

- Looking at the units (of measurement); *[1 mark]*
- Interpreting the context of the quantity; *[1 mark]*
- Using units to check the homogeneity of physical equations; *[1 mark]*

***[Total: 1 mark]***

## Question 2 Easy [6 marks]

### Part (a) [3 marks]

Complete Table 1.1 to show the name and symbol of each physical quantity and the corresponding unit.

**Table 1.1**

| physical quantity | symbol | unit |
|---|---|---|
| velocity | $v$ | m $s^{-1}$ |
| | $W$ | N m |
| stress | $\sigma$ | |
| | | kg $m^{-3}$ |

**Mark Scheme (3 marks):**

The correctly completed table is as follows:

| physical quantity | symbol | unit |
|---|---|---|
| velocity | $v$ | m $s^{-1}$ |
| ***work done*** | $W$ | N m |
| stress | $\sigma$ | $N\ m^{-2}$ |
| *density* | $\rho$ | kg $m^{-3}$ |

- One mark for each correct row:
  - work done *[1 mark]*
  - N $m^{-2}$ *[1 mark]*

- density **AND** $\rho$ *[1 mark]*

***[Total: 3 marks]***

**Part (b)** [3 marks]

Calculate a reasonable estimate for the work done by a person lifting a dining room chair 1.0 m above the ground.

**Mark Scheme (3 marks):**

Estimate the work done by an average adult person lifting a dining room chair 1.0 m above the ground:

Estimate the average mass of a dining room chair:

- Mass of chair = 1 kg - 10 kg *[1 mark]*

Calculate an estimate of the work done, *W*:

- $W = Fd = mgd$ *[1 mark]*
- Range: $W$ = 10 J - 100 J *[1 mark]*
- Example calculation:
  - $W = mgd$
  - Where $m \approx 5 \text{ kg}$, $g \approx 10 \text{ N kg}^{-1}$, and $d = 1.0 \text{ m}$
  - $W \approx 5 \times 10 \times 1.0 \approx 50 \text{ J}$

***[Total: 3 marks]***

*In an estimation answer, a range of estimated values will be acceptable.*

## Question 3 Easy [9 marks]

### Part (a) [3 marks]

A physics student is investigating the properties of materials. She has a stack of 500 sheets of paper and needs to determine the thickness of a single sheet.

(i) Describe a method she could use to find the thickness of one sheet of paper without using a micrometer.

[2]

(ii) State an appropriate unit for the thickness of one sheet of paper.

[1]

**Mark Scheme (3 marks):**

(i) To find the thickness of one sheet of paper without using a micrometer:

- Use a ruler to measure the total thickness (of the stack); *[1 mark]*
- Divide the measurement by 500; *[1 mark]*

(ii) A suitable unit for the thickness of paper is:

- millimetres / mm

  **OR**

- micrometres / μm; *[1 mark]*

***[Total: 3 marks]***

### Part (b) [3 marks]

The student then investigates simple harmonic motion using a mass-spring system. She wishes to determine the period of oscillation of the system.

Describe a method the student could use to obtain an accurate value for the period of one complete oscillation.

**Mark Scheme (3 marks):**

A method the student could use to obtain an accurate value for the period of one complete oscillation:

- Measure the time for 10 complete oscillations; *[1 mark]*
- Divide the total time by 10 to find the period; *[1 mark]*
- Repeat the measurement several times and calculate a mean; *[1 mark]*

***[Total: 3 marks]***

**Part (c)** [1 mark]

During a break in her experiments, the student scrunches up a sheet of paper and throws it horizontally from shoulder height across the laboratory.

Estimate the horizontal distance travelled by the paper ball before it hits the floor.

**Mark Scheme (1 marks):**

The paper ball is thrown:

- 2 m *[1 mark]*

***[Total: 1 mark]***

*Any value between 1 and 5 m is acceptable to gain the marks in this question.*

**Part (d)** [2 marks]

The student then throws a tennis ball of the same size using the same technique from the same height.

State whether the tennis ball travels a greater horizontal distance than the paper ball. Explain your answer.

**Mark Scheme (2 marks):**

The tennis ball will:

- Travel further (than the paper ball); *[1 mark]*

Explanation:

- The tennis ball has a greater mass, so it experiences a smaller deceleration due to air resistance; *[1 mark]*

***[Total: 2 marks]***

## Question 4 Medium [12 marks]

### Part (a) [2 marks]

Estimate the speed of an apple as it hits the ground after falling from a tree.

**Mark Scheme (2 marks):**

Estimate the speed of an apple as it hits the ground after falling from a tree:

- Height of apple on a tree, $h \approx 5$ m *[1 mark]*

*Acceptable range for h = 2 to 20 m*

Use energy conservation to estimate the speed:

- Change in GPE = Gain in KE
- $mgh = \frac{1}{2}mv^2$
- $v^2 = 2gh \quad \Rightarrow \quad v = \sqrt{2gh}$
- $v \approx \sqrt{2 \times 9.81 \times 5}$
- $v \approx 9.91 \approx 10$ m s$^{-1}$ *[1 mark]*

*Acceptable range for v = 6 to 20 m* s$^{-1}$

***[Total: 2 marks]***

### Part (b) [2 marks]

Estimate the number of kilowatt-hours used by a lightbulb if switched on for a whole day.

**Mark Scheme (2 marks):**

Estimate the number of kilowatt-hours used by a lightbulb if switched on for a whole day:

- Power output of a lightbulb ≈ 70 W *[1 mark]*

*Acceptable range for power = 20 to 100 W*

Convert from W to kWh in 24 hours:

- $P = \frac{E}{t} \quad \Rightarrow \quad E = Pt$
- Kilowatt-hours is a unit of energy, so $EkWh = \frac{PW \times thr}{1000}$
- $EkWh \approx \frac{70 \times 24}{1000}$
- $E$(kWh) ≈ 1.68 ≈ 1.7 kWh per day *[1 mark]*

*Acceptable range for E(kWh) = 0.5 to 2.4 kWh*

***[Total: 2 marks]***

**Part (c)** **[4 marks]**

Estimate the density, in g $cm^{-3}$, of the head of an adult person.

**Mark Scheme (4 marks):**

Estimate the density, in g $cm^{-3}$, of the head of an adult person:

- Radius of a head, $r \approx 10$ cm *[1 mark]*

*Acceptable range for radius = 7 to 12 cm*

- Mass of a head, $m \approx 5$ kg = 5000 g *[1 mark]*

*Acceptable range for mass = 1 to 10 kg (1000 to 10 000 g)*

Use the equation for volume of a sphere with the density equation:

- Volume of an adult's head ≈ volume of a sphere
- Volume of a sphere: $V = \frac{4}{3}\pi r^3$
- Density: $\rho = \frac{m}{V} = \frac{m}{\frac{4}{3}\pi r^3}$
- $\rho = \frac{5000}{\frac{4}{3}\pi \times 10^3}$ *[1 mark]*
- $\rho = 1.194 = 1.2$ g $cm^{-3}$ *[1 mark]*

*Acceptable range for density = 0.1 to 7.0 g $cm^{-3}$*

***[Total: 4 marks]***

*When asked a question, such as this, where you likely have no reference to make a reasonable estimate for something, like the mass of a person's head, think about reasonable approximations you could make*

*For example, if the mass of an adult person is about 70 kg, you could make a reasonable assumption that their head might make up about 10% of this mass, which would be 7 kg*

*It's important that you communicate this in your answer so the examiner can see your lines of thought, and even if you are outside of the acceptable range, you could still earn extra marks for method and reasoning*

**Part (d)** **[4 marks]**

The radius of the Earth is approximately $6.4 \times 10^6$ m.

Estimate the number of people it would take to circle the Earth holding hands.

**Mark Scheme (4 marks):**

Estimate the number of people it would take to circle the Earth holding hands:

- Radius of Earth, $r = 6.4 \times 10^6$ m
- Arm span ≈ 2 m *[1 mark]*

*Acceptable range for arm span estimate = 1 to 3 m*

Calculate the circumference of Earth:

- Circumference of a circle = $2\pi r$
- Circumference of Earth = $2\pi \times (6.4 \times 10^6) = 4 \times 10^7$ m *[1 mark]*

Calculate how many people would be needed:

- Each person spans approximately 2 m
- Number of people ≈ $\frac{circumference\ \ of\ \ Earth}{arm\ \ span}$
- Number of people ≈ $\frac{4 \times 10^7}{2}$ *[1 mark]*
- Number of people ≈ $2 \times 10^7$ *[1 mark]*

*Acceptable range for number of people estimate = $1.3 \times 10^7$ to $4 \times 10^7$*

***[Total: 4 marks]***

*If you have approximated the arm span of a person to a different value than 2 m, that's ok, as long as it is a **reasonable** approximation. You may have said, 1.5 m or somewhere in between, and that's ok. You would still get the marks. But, if you approximated it to 5 m, this is **not** a reasonable estimation and you would not get the first mark. However, you would still get the other two marks if your calculations were correct, no matter what number you used.*

*Questions that ask you to **estimate** may not always contain an unknown value that you need to approximate, like arm span in this question. They may just use inexact values such as the diameter of the Earth, or the mass of a proton as 1 atomic mass unit. Don't let this throw you, the calculation is just the same as it would be if you were using exact values. If you do approximate something, like the hand span, just remember to use the 'approximately equal to' sign ≈*

## Question 5 Medium [12 marks]

### Part (a) [3 marks]

X–ray pulsars are detected by X–ray telescopes on a satellite in low Earth orbit 2000 km above the surface of the Earth.

Calculate the number of cubic millimetres ($mm^3$) in a volume of 2000 $km^3$.

**Mark Scheme (3 marks):**

List known quantities:

- Volume = 2000 $km^3$

Recall the metric multipliers:

- k = kilo = $10^3$
- m = milli = $10^{-3}$

Convert $km^3$ into $mm^3$:

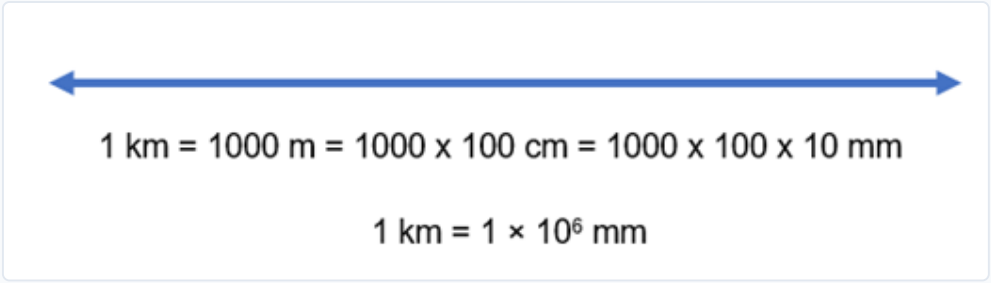
- 1 km = $1 \times 10^6$ mm *[1 mark]*
- 1 $km^3$ = $(1 \times 10^6)^3$ $mm^3$ = $1 \times 10^{18}$ $mm^3$ *[1 mark]*

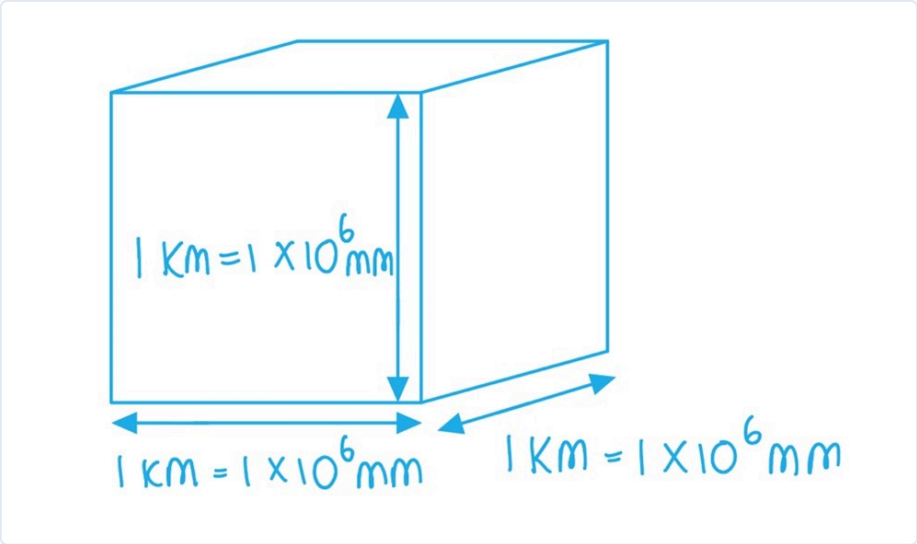
Convert 2000 $km^3$ into $mm^3$:

- 2000 $km^3$ = $2000 \times (1 \times 10^{18}) = 2 \times 10^{21}$ $mm^3$ *[1 mark]*

***[Total: 3 marks]***

*Recognising that there are $1 \times 10^{18}$ $mm^3$ in 1 $km^3$ is the key to answering this question correctly.*






*Volume of a cube = length × width × height = (1 × $10^6$) × (1 × $10^6$) × (1 × $10^6$) = 1 × $10^{18}$ $mm^3$*

**Part (b)** **[3 marks]**

X–rays from a pulsar travel at the speed of light and are detected on Earth with a wavelength of 8.0 nm.

Calculate the frequency of the X–rays. Give your answer to an appropriate number of significant figures.

**Mark Scheme (3 marks):**

List the known quantities and expand the multipliers:

- Wavelength of pulsar = 8.0 nm = 8.0 × $10^{-9}$ m *[1 mark]*

From the data page:

- Speed of light in free space, $c$ = 3.00 × $10^8$ m $s^{-1}$

Recall the metric multiplier:

- Nano = × $10^{-9}$

Recall and rearrange the wave speed equation:

- $c = f\lambda$
- $f = \frac{c}{\lambda}$ *[1 mark]*

Calculate the frequency:

- $f = \frac{3.0 \times 10^8}{8.0 \times 10^{-9}}$ = 3.75 × $10^{16}$ Hz *[1 mark]*

Write answer to an appropriate number of significant figures:

- $f$ = 3.8 × $10^{16}$ Hz *[1 mark]*

***[Total: 3 marks]***

*Make sure you read the question correctly, so you write your final answer in the correct form. 8.0 nm is the only quantity in the question. This is written to 2 significant figures. So, the final answer must also be written to 2 significant figures.*

**Part (c)** [3 marks]

X–rays from the nearest pulsar PSR J0109–1431 take 8.82 Gs to travel to Earth.

Calculate the number of oscillations of the X–rays from the pulsar to the surface of the Earth. Give your answer to an appropriate number of significant figures.

**Mark Scheme (3 marks):**

List known quantities and expand the multipliers:

- Time from pulsar to Earth = 8.82 Gs = $8.82 \times 10^9$ s
- Frequency, $f$ = $3.75 \times 10^{16}$ Hz (from part (b))

Calculate the time period for one oscillation:

- $T = \frac{1}{f}$
- $T = \frac{1}{3.75 \times 10^{16}}$ *[1 mark]*

Find the number of oscillations:

- Number of oscillations = $\frac{\text{total time of journey}}{\text{time for 1 oscillation}}$
- Number of oscillations = $3.75 \times 10^{16} \times 8.82 \times 10^9$ = $3.3075 \times 10^{26}$ *[1 mark]*

State to an appropriate number of significant figures:

- Number of oscillations = $3.31 \times 10^{26}$ (3 s.f) *[1 mark]*

***[Total: 3 marks]***

*Use the metric multipliers to correctly calculate the value of the giga prefix in standard form. Use your knowledge of time period and frequency to find the time for one oscillation of the pulsar. 8.82 Gs is the only quantity given in the question. This is to 3 significant figures, so the final answer must also be to 3 significant figures.*

**Part (d)** [3 marks]

Show the distance from Earth to PSR J0109–1431 is around 280 light years.

**Mark Scheme (3 marks):**

To show that the distance is around 280 light years:

List known quantities:

- Time = 8.82 Gs = $8.82 \times 10^9$ s

- Distance = 280 light years

From the data page:

- Speed of light in free space, $c$ = 3.00 × $10^8$ m $s^{-1}$

Calculate the distance in m:

- Distance = speed of light × time
- Distance = (3 × $10^8$) × (8.82 × $10^9$) = 2.646 × $10^{18}$ m *[1 mark]*

Convert distance in m to light years:

- 1 light year = c × 1 year = $3.00 \times 10^8 \times 60 \times 60 \times 24 \times 365 = 9.46 \times 10^{15}$ m *[1 mark]*

Divide the total distance by the distance in one light year

- $\frac{2.646 \times 10^{18}}{9.46 \times 10^{15}}$ = 279.704 = 280 light years *[1 mark]*

*[Total: 3 marks]*

*Don't forget that light years are a unit of distance not time.*

*Remember to use previous parts of the question for quantities that appear missing from the question.*

## Question 6 Medium [8 marks]

### Part (a) [1 mark]

Define the term *physical quantity*.

**Mark Scheme (1 marks):**

A physical quantity is...

- A physical property that can be measured and quantified; *[1 mark]*

***[Total: 1 mark]***

*An example of a physical quantity is mass which is measured in kg.*

### Part (b) [3 marks]

Physical quantities have symbols and units. Indicate, with a tick (✓), which of the following are **not** valid physical quantities.

| | |
|---|---|
| kinetic energy $E_k$ measured in kg $m^2$ $s^{-2}$ | ☐ |
| pressure $P$ measured in N $kg^{-1}$ | ☐ |
| Young modulus $E$ measured in Pa | ☐ |
| frequency $f$ measured in s | ☐ |
| specific heat capacity $c$ measured in J $kg^{-1}$ | ☐ |
| weight $W$ measured in kg m $s^{-2}$ | ☐ |

**Mark Scheme (3 marks):**

The quantities which are not valid physical quantities are:

| | |
|---|---|
| kinetic energy $E_k$ measured in kg $m^2$ $s^{-2}$ | |
| pressure $P$ measured in N $kg^{-1}$ | ✓ |

| | |
|---|---|
| Young modulus $E$ measured in Pa | |
| frequency $f$ measured in s | ✓ |
| specific heat capacity $c$ measured in J $kg^{-1}$ | ✓ |
| weight $W$ measured in kg m $s^{-2}$ | |

- pressure $P$ measured in N $kg^{-1}$; *[1 mark]*
- frequency $f$ measured in s; *[1 mark]*
- specific heat capacity $c$ measured in J $kg^{-1}$; *[1 mark]*

***[Total: 3 marks]***

*The correct physical quantities are:*

- *$E_k$ = kinetic energy, measured in J or kg $m^2$ $s^{-2}$*
- *$P$ = pressure, measured in Pa or N $m^{-2}$*
- *$E$ = Young modulus, measured in Pa or N $m^{-2}$*
- *$f$ = frequency in Hz or $s^{-1}$*
- *$c$ = specific heat capacity, measured in J $kg^{-1}$ $K^{-1}$*
- *$W$ = weight measured in N or kg m $s^{-2}$*

*You do not need to memorise all the symbols and units listed in the specification, but you do need a good level of recognition and understanding of which units belong to which quantity. Knowing the units will help you with the maths in physics, because the units tell you some of the physical quantities that you need in the calculation. For example the units for latent heat, L, are joules per kilogram, so you know you need energy divided by mass in that calculation.*

*In other courses and textbooks, you may find different symbols used for different quantities. It is important that you use the symbols given in the CIE IAL specification.*

**Part (c)** **[4 marks]**

Complete the table providing reasonable estimates for some physical quantities.

| quantity | estimated magnitude |
|---|---|
| Weight of an adult human | |

| | |
|---|---|
| | 3600 s |
| Wavelength of green light | |
| | 20 °C |

**Mark Scheme (4 marks):**

A table scoring 4 marks:

| quantity | estimated magnitude |
|---|---|
| Weight of an adult human | ***700 N*** *(±100 N) [1 mark]* |
| ***Seconds in an hour*** *[1 mark]* | 3600 s |
| Wavelength of green light | ***550 nm*** **OR** ***550×10$^{-9}$ m*** *[1 mark]* |
| ***Room temperature*** *[1 mark]* | 20 °C |

***[Total: 4 marks]***

*The CIE IAL specification states that you should be able to make reasonable estimates of physical quantities included within the syllabus. The ones in this question are common ones.*

*To estimate you can use the nearest order of magnitude, or choose rounded numbers to use in your calculation. For example, an estimate for the mass of an adult human is 70 kg, an estimate for the acceleration of freefall is 10 m s$^{-2}$, therefore an estimate of weight is 70 × 10 = 700 N.*

## Question 7 Hard [4 marks]

### Part (a) [4 marks]

Complete Table 1.1 to show the SI base unit of each physical quantity and a reasonable estimate of the order of magnitude.

**Table 1.1**

| physical quantity | SI base unit | order of magnitude |
|---|---|---|
| acceleration of free-fall on Earth ($g$) | m $s^{-2}$ | |
| Stefan-Boltzmann constant ($\sigma$) | | $10^{-7}$ |
| speed of a $\beta$ particle | | |
| specific heat capacity of water ($c$) | | |

**Mark Scheme (4 marks):**

The correct SI base units and estimates for order of magnitude are:

***One*** *mark for each correct row:*

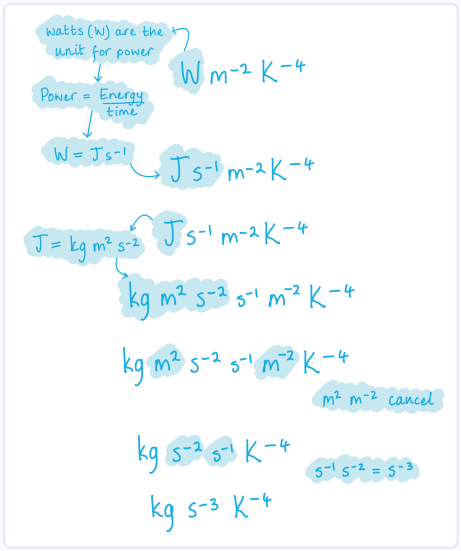
| physical quantity | SI base unit | order of magnitude |
|---|---|---|
| acceleration of free-fall on Earth ($g$) | m $s^{-2}$ | $\boldsymbol{10^{1}}$ |
| Stefan-Boltzmann constant ($\sigma$) | $\boldsymbol{kg\ s^{-3}\ K^{-4}}$ | $10^{-7}$ |
| speed of a $\beta$ particle | $\boldsymbol{m\ s^{-1}}$ | $\boldsymbol{10^{8}}$ |
| specific heat capacity of water ($c$) | $\boldsymbol{m^{2}\ s^{-2}\ K^{-1}}$ | $\boldsymbol{10^{3}}$ |

***[Total: 4 marks]***

*The acceleration of freefall on Earth, g, is 9.81 m $s^{-2}$. This quantity is already given in SI base units and the order of magnitude is $10^{1}$ as $10^{1}$ = 10.*

*The Stefan-Boltzmann constant σ is given in the data booklet as 5.67 × $10^{-8}$ W $m^{-2}$ $K^{-4}$. Rounded to the nearest power of ten, the order of magnitude is $10^{-7}$*

*The units of the Stefan-Boltzmann constant σ are:*

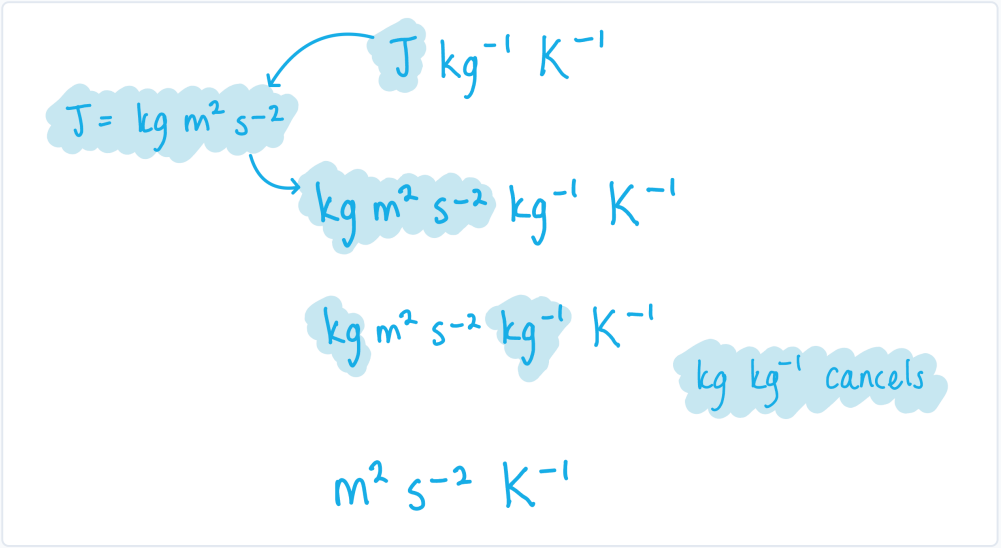



*The speed of a β particle can be approximated to the speed of light since a β particle is a high-speed electron. The speed of light in a vacuum c is given in the data booklet as 3.00 × $10^{8}$ m $s^{-1}$*

*Metres and seconds are already SI base units, and rounded to the nearest power of 10, the order of magnitude is $10^{8}$*

*The specific heat capacity c of water is not given in the data booklet, but you should be familiar with it from practising calculations. The specific heat capacity of water is about 4200 J $kg^{-1}$ $K^{-1}$, or 4.2 × $10^{3}$ J $kg^{-1}$ $K^{-1}$*

*Rounded to the nearest power of 10, the order of magnitude is $10^{3}$ and the units are:*



## Question 8 Hard [4 marks]

### Part (a) [4 marks]

Complete Table 2.1 to show a reasonable estimate of the order of magnitude of each physical quantity and the corresponding unit.

**Table 2.1**

| quantity | order of magnitude | unit |
|---|---|---|
| mass of a car | | kg |
| thickness of a sheet of paper | | mm |
| density of air (at room temperature and pressure) | | $g\ cm^{-3}$ |
| time between two heartbeats | | s |

**Mark Scheme (4 marks):**

Estimates of the orders of magnitude of the quantities are:

| quantity | order of magnitude | unit |
|---|---|---|
| mass of a car | $1 \times 10^{3}$ *[1 mark]* | kg |
| thickness of a sheet of paper | $1 \times 10^{-1}$ *[1 mark]* | mm |
| density of air (at room temperature and pressure) | $1 \times 10^{-3}$ *[1 mark]* | $g\ cm^{-3}$ |
| time between two heartbeats | $1 \times 10^{0}$ *[1 mark]* | s |

***[Total: 4 marks]***

*You should know that the mass of a car is of the order of 1 ton which is 1000 kg.*

*You can estimate the thickness of a sheet of paper by considering a stack of 500 sheets (think about the stacks of paper that you can buy for printers), these are about 1 inch thick which is 2.5 cm or 25 mm, so 1 sheet of paper is about 25 / 500 = 0.05 mm thick.*

*You should also know that the density of air is about 1 kg $m^{-3}$ or 0.001 g $cm^{-3}$.*

*You can consider a heartbeat and the number of beats per minute to calculate the time between two beats, it's roughly 1 s depending on your resting heart rate.*